Supplement (Extension): Epistemic Limits and the Loss-Tracing Tool (LTT)

Operational formalization of mismatch, loss channels, and implementable algorithms

This document is an extension to the anchor paper: "Two-Substrate Theory and the Genesis–Guidance–Transformation–Void Cycle"

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Abstract

This supplement extends the anchor paper by formalizing (i) the epistemic assumptions implicit in contemporary physics, (ii) the "observational trap" created by particle–field language, and (iii) a concrete, computable *Loss-Tracing Tool (LTT)*. The LTT is defined as a projection-and-residual pipeline: select a common observable space, project matter-side and energy/field-side descriptions into that space, compute a calibrated mismatch metric, decompose the mismatch into loss channels, and accumulate loss into one or more Void ledgers (energetic and informational). The specification includes validity conditions, calibration requirements, and algorithms suitable for simulations and experimental residual analysis. Dark-sector interpretations are stated as *hypotheses* and are explicitly separated from the operational tool.

Contents

1 Scope and Relation to the Anchor Paper **3**

2 Epistemic Foundations **3**

2.1 UTC tri-partite reality 3

2.2 The hidden axiom and its boundary 3

2.3 Model vs ontology (the discipline) 3

3 The Observational Trap **4**

4 Operational Ontology (for the Tool) **4**

4.1 Three bookkeeping categories 4

5 The Loss-Tracing Tool (LTT): Formal Specification **5**

5.1 Common observable space 5

5.2 Two projections (matter-side vs energy/field-side) 5

5.3 Residual and calibrated mismatch 5
5.4 Channelization (energy-like vs information-like loss) 5
5.5 Void ledgers (energetic and informational) 6

6 Tool Axioms (Validity Conditions) 6

7 Metric Choices (Continuous, Counted, Quantum) 6
7.1 Gaussian / Mahalanobis geometry 6
7.2 Poisson / likelihood geometry for event counts 7
7.3 Quantum-state distance (optional module) 7

8 Algorithmic Design: Implementable LTT Pipeline 7
8.1 Core pipeline (time-series or event stream) 7
8.2 Residual structure modules (what makes LTT useful) 7

9 Validation Protocol (Make It Experimental) 8
9.1 Minimum falsifiability checklist 8

10 Application Templates (Operational) 8
10.1 Template A: Laboratory energy accounting (closure demonstration) 8
10.2 Template B: Event-based detectors (direct detection style) 8
10.3 Template C: Astrophysical residual maps (missing-component inference) 8

11 Dark-Sector Interpretations (Explicitly Hypotheses) 9

A Symbol Summary (Supplement) 9

1 Scope and Relation to the Anchor Paper

This document does **not** rewrite the anchor paper. It extends it in three ways:

1. It formalizes an **epistemic layer**: what counts as "real" is constrained by what can be operationally measured and expressed in the available ontology.
2. It introduces a precise statement of the **observational trap**: signals are forced into particle/field interpretations by the language and apparatus of measurement.
3. It specifies a computable **Loss-Tracing Tool (LTT)** that turns "mismatch" into a measurable object, decomposed into channels, with calibrated uncertainty.

Remark 1.1. In this supplement, the word "Void" is used as a **ledger substrate**—a bookkeeping layer for mismatch that remains after projection. Whether this ledger corresponds to a new physical entity is treated as a separate hypothesis, not a premise of the tool.

2 Epistemic Foundations

2.1 UTC tri-partite reality

Postulate 2.1 (UTC tri-partite reality)**.** We distinguish three layers:

- **R1 (subjective):** private experience (qualia), not directly formalizable.
- **R2 (intersubjective):** language, symbols, mathematics, models; formalization lives here.
- **R3 (objective):** reality independent of belief, accessible only via traces in data.

Postulate 2.2 (Observables-first principle)**.** A formal claim is meaningful only when it is tied to a procedure that maps potential measurements into a well-defined mathematical object.

2.2 The hidden axiom and its boundary

Postulate 2.3 (Particle–Field axiom)**.** Within mainstream physics, existence is operationally identified with describability as either: (i) a particle-like localized excitation, or (ii) a field-like continuous distribution. Measurement outcomes are interpreted through this ontology.

Remark 2.4. Theorem 2.3 is not asserted as false; it is asserted as *a language boundary*. A language boundary can create systematic misclassification when phenomena fall outside the ontology but still leave measurable traces.

2.3 Model vs ontology (the discipline)

Definition 2.5 (Model contract)**.** A **model contract** is the explicit triple:

$$\text{(measurement protocol)} \;\Rightarrow\; \text{(observable definition)} \;\Rightarrow\; \text{(comparison rule)}.$$

The LTT is a formalization of the comparison rule, and it is valid only under a specified contract.

3 The Observational Trap

Definition 3.1 (Observational trap)**.** To detect an entity, one must interact with it. Interaction produces signals in the detector. Signals are interpreted within the available ontology. If an unknown cause produces energy deposition indistinguishable from a particle-like event under the existing model contract, then the apparatus and language will label it "particle evidence" even if the underlying cause is not particle-like.

Observational Trap – Schematic

$$\text{Unknown cause } X \xrightarrow{\text{interaction with detector}} \text{signal } S$$

$$\text{Interpretation inside particle–field ontology: } S \Rightarrow \text{particle/field label}$$

$$\text{But: } X \text{ may be a different kind of energy structure (under a broader ontology).}$$

Figure 1: Signals are automatically interpreted within the available model language. The LTT is designed to quantify and structure the residual.

Remark 3.2. The goal of LTT is not to "prove new physics." The goal is to turn the trap into a measurable residual: what remains unexplained *under an explicit model contract.*

4 Operational Ontology (for the Tool)

4.1 Three bookkeeping categories

Definition 4.1 (Space substrate (matter-side))**.** Space $\mathcal{X}$ is the domain in which matter has position. Matter at time t can be represented as

$$\mathcal{M}(t) = \{(\boldsymbol{x}_i(t), m_i, q_i, \boldsymbol{s}_i, r_i)\}_{i=1}^{N(t)}.$$

Definition 4.2 (Time substrate (energy/field-side))**.** Energy is represented as fields or processes parameterized by time:

$$E : \mathcal{X} \times \mathbb{R} \to \mathbb{R}^k,$$

together with any external driving inputs.

Definition 4.3 (Void ledger substrate)**.** The Void $\mathcal{V}$ is defined operationally as a **ledger of mismatch** after projection. In this supplement we explicitly separate:

$$\mathcal{V}_E \quad \text{(energetic ledger, J-like)}, \qquad \mathcal{V}_I \quad \text{(informational ledger, nats/bits-like)}.$$

Remark 4.4. The separation $\mathcal{V}_E, \mathcal{V}_I$ prevents unit-mixing. Whether $\mathcal{V}_E$ and $\mathcal{V}_I$ correspond to one deeper entity is a hypothesis, not required for the tool.

5 The Loss-Tracing Tool (LTT): Formal Specification

5.1 Common observable space

Definition 5.1 (Observable space)**.** Let $\mathcal{O} \subseteq \mathbb{R}^n$ be a space of **observables**. Each component is a calibrated measurement or derived statistic (e.g., energy deposition in channels, count spectra, trajectories, shear estimates, etc.). The common reference system is identified with $\Sigma_{\text{common}} \equiv \mathcal{O}$.

5.2 Two projections (matter-side vs energy/field-side)

Definition 5.2 (Reference systems)**.** Σ_X denotes the matter-side reference system (states describing matter in $\mathcal{X}$). Σ_T denotes the energy/field-side reference system (states describing energy/fields and inputs over t).

Definition 5.3 (Projection operators)**.** Define two projections into the same $\mathcal{O}$:

$$\Pi_X : \Sigma_X \to \mathcal{O}, \qquad \Pi_T : \Sigma_T \to \mathcal{O}.$$

These are not merely symbols: each implies a calibration procedure mapping theoretical variables into predicted observables.

5.3 Residual and calibrated mismatch

Definition 5.4 (Residual (mismatch vector))**.** For matter-side state $A \in \Sigma_X$ and energy/field-side state $B \in \Sigma_T$,

$$r(A, B) := \Pi_X(A) - \Pi_T(B) \in \mathbb{R}^n.$$

Definition 5.5 (Calibrated mismatch metric)**.** Let $\mathbf{W} \succeq 0$ be a weight matrix. Define

$$\delta(A, B) := \sqrt{r(A, B)^\top \mathbf{W}\, r(A, B)}.$$

A canonical choice is $\mathbf{W} = \mathbf{\Sigma}^{-1}$, where $\mathbf{\Sigma}$ is the covariance of measurement noise and model uncertainty in $\mathcal{O}$.

Remark 5.6 (Why calibration is non-negotiable). If δ is computed without $\mathbf{W}$, the result depends on units and arbitrary scaling. With $\mathbf{W} = \mathbf{\Sigma}^{-1}$, δ becomes dimensionless and corresponds to statistical significance under the model contract.

5.4 Channelization (energy-like vs information-like loss)

Definition 5.7 (Loss channel vector)**.** Define loss channels as a vector

$$\mathbf{L} = (L^{(1)}, \ldots, L^{(m)}) \in \mathbb{R}^m,$$

representing distinct mechanisms: heat, radiation, unmodeled leakage, entropy production proxies, coherence loss proxies, etc.

Definition 5.8 (Channelization map)**.** A channelization map

$$\mathcal{C} : \mathbb{R}^n \to \mathbb{R}^m, \qquad \mathbf{L} = \mathcal{C}(r)$$

decomposes residual structure into channels. $\mathcal{C}$ may be: (i) linear projection onto a chosen basis, or (ii) nonlinear inference (e.g., a fitted residual model), but must be defined under the contract.

5.5 Void ledgers (energetic and informational)

Definition 5.9 (Energetic ledger update)**.** Define an energetic-loss extractor $h_E : \mathbb{R}^m \to \mathbb{R}_{\geq 0}$ and accumulate:

$$V_E(t) = V_E(0) + \int_0^t h_E(\mathbf{L}(\tau))\ \mathrm{d}\tau.$$

Definition 5.10 (Informational ledger update)**.** Define an information-loss extractor $h_I : \mathbb{R}^m \to \mathbb{R}_{\geq 0}$ and accumulate:

$$V_I(t) = V_I(0) + \int_0^t h_I(\mathbf{L}(\tau))\ \mathrm{d}\tau.$$

Remark 5.11. In legacy notation, one may write $E_{\mathcal{V}}(t)$ as a scalar if only the energetic ledger is tracked. This supplement recommends (V_E, V_I) as the default.

6 Tool Axioms (Validity Conditions)

Axiom 6.1 (Well-definedness)**.** Π_X and Π_T must map into the same $\mathcal{O}$, and each component of $\mathcal{O}$ must have a defined calibration protocol.

Axiom 6.2 (Dimensional consistency)**.** Either (i) observables are dimensionless, or (ii) $\mathbf{W}$ must normalize units so that δ is dimensionless.

Axiom 6.3 (Closure test)**.** In a calibration scenario where all relevant channels are measured and modeled, the tool must satisfy:

$$\mathbb{E}[\mathbf{L}] \approx \mathbf{0} \quad \text{within uncertainty.}$$

Persistent nonzero residual implies missing channel accounting or a faulty projection contract.

Axiom 6.4 (Identifiability vs model error)**.** The tool must distinguish genuine loss from systematic model error. Practically, this requires nuisance parameters and uncertainty propagation so that structured residuals are not misclassified as "Void" by default.

Remark 6.5. Theorems 6.3 and 6.4 are what keep Void from becoming a non-falsifiable garbage bin.

7 Metric Choices (Continuous, Counted, Quantum)

7.1 Gaussian / Mahalanobis geometry

If r is approximately Gaussian under the contract, $\mathbf{W} = \mathbf{\Sigma}^{-1}$ is appropriate and δ^2 behaves like a chi-square statistic.

7.2 Poisson / likelihood geometry for event counts

When observables are counts k_b in bins with predicted means μ_b, use a deviance distance:

$$\delta^2_{\text{Pois}} := 2 \sum_b \left(\mu_b - k_b + k_b \log \frac{k_b}{\mu_b} \right).$$

This is LTT-consistent with a likelihood-based geometry.

7.3 Quantum-state distance (optional module)

If the observable is a density matrix ρ, mismatch can be measured via distances such as trace distance or relative entropy. This is optional and should be introduced only under an explicit measurement contract.

8 Algorithmic Design: Implementable LTT Pipeline

8.1 Core pipeline (time-series or event stream)

Algorithm 1: Loss-Tracing Tool (LTT): Core Pipeline

Input: Data stream $\mathcal{D}_k$ (time steps or events); observable constructor $g : \mathcal{D} \to \mathcal{O}$; model states $A_k \in \Sigma_X$, $B_k \in \Sigma_T$; projections Π_X, Π_T; calibration model for $\mathbf{W}$ (or likelihood geometry); channelization map $\mathcal{C}$; ledger extractors h_E, h_I; decision policy $\tau(\cdot)$.

Output: Residuals r_k, mismatch δ_k, channels $\mathbf{L}_k$, ledgers V_E, V_I, anomaly flags.

Init *Set $V_E \leftarrow V_E(0)$, $V_I \leftarrow V_I(0)$. Initialize $\mathbf{W}$ from calibration/control data (or set likelihood parameters).*

foreach *index k* **do**

  Build observed observable $y_k \leftarrow g(\mathcal{D}_k)$.

  Predict observables $\hat{y}_k^{(X)} \leftarrow \Pi_X(A_k)$, $\hat{y}_k^{(T)} \leftarrow \Pi_T(B_k)$.

  Residual $r_k \leftarrow \hat{y}_k^{(X)} - \hat{y}_k^{(T)}$.

  ▷ `Mismatch metric`

  Compute $\delta_k \leftarrow \sqrt{r_k^\top \mathbf{W}\, r_k}$ **or** $\delta_k \leftarrow \delta_{\text{likelihood}}(y_k, \hat{y}_k)$.

  ▷ `Channelization and ledger update`

  $\mathbf{L}_k \leftarrow \mathcal{C}(r_k)$.

  $V_E \leftarrow V_E + h_E(\mathbf{L}_k)\,\Delta t$.

  $V_I \leftarrow V_I + h_I(\mathbf{L}_k)\,\Delta t$.

  ▷ `Decision / flagging`

  Flag anomaly if $\delta_k > \tau(k)$ or if structural tests trigger.

Return $\{r_k, \delta_k, \mathbf{L}_k, V_E, V_I\}$ and flags.

8.2 Residual structure modules (what makes LTT useful)

A single large residual can be noise. The tool becomes powerful when residuals show structure: periodicity, cross-detector coherence, narrow spectral features, spatial morphology.

Algorithm 2: Residual-Structure Plug-ins (Optional)

Input: Residual stream r_k, mismatch δ_k, metadata (time, location, detector ID).
Output: Structure scores and candidate explanations (instrumental vs physical).

Compute autocorrelation / periodogram on selected residual components; score periodic peaks.
If multiple detectors: compute cross-correlation / coherence on aligned windows.
Run change-point detection on δ_k (regime shifts).
In energy-binned residuals: search for narrow lines above smooth backgrounds.
Return structure scores; map to hypotheses under the contract.

9 Validation Protocol (Make It Experimental)

9.1 Minimum falsifiability checklist

To claim LTT "works" under a contract, require:

1. **Calibration closure:** pass Theorem 6.3 in controlled conditions.
2. **Intervention response:** change a known parameter (shielding, temperature, drive) and predict the sign/direction of δ and channel changes.
3. **Cross-validation:** fit nuisance parameters on one dataset, test on another without retuning the core projection.
4. **Systematics audit:** show residual structure is not explainable by known nuisance models.

Remark 9.1. This protocol prevents "Void" from becoming a metaphysical claim. Void is accepted only as a residual ledger after audits and interventions.

10 Application Templates (Operational)

10.1 Template A: Laboratory energy accounting (closure demonstration)

Example 10.1 (Energy budget closure). Let $\mathcal{O} = (E_{\text{in}}, E_{\text{mech}}, E_{\text{heat}}, E_{\text{rad}})$. Under full instrumentation, $\mathcal{C}(r)$ should yield near-zero channels (closure). Remove one sensor (e.g., heat) and the residual should move into the corresponding channel with predictable dynamics.

10.2 Template B: Event-based detectors (direct detection style)

Example 10.2 (Count-spectrum residual). Let $\mathcal{O}$ be binned counts k_b and auxiliary discriminants. Π_T generates predicted means μ_b from background + signal model. Mismatch uses deviance δ^2_{Pois}. Channelization separates ER/NR-like components, spatial nonuniformity, time dependence, etc.

10.3 Template C: Astrophysical residual maps (missing-component inference)

Example 10.3 (Residual field mapping). Let $\mathcal{O}$ be lensing shear samples or rotation-curve points. Π_T predicts observables from visible matter under a baseline gravity model. The residual is

converted into a "missing component" map via a reconstruction operator $\mathcal{R}$. LTT supplies calibrated significance and stability under systematics variations.

11 Dark-Sector Interpretations (Explicitly Hypotheses)

Hypothesis 11.1 (Dark-sector as structured residual under a baseline contract)**.** A significant portion of what is called "dark" may be operationally viewed as structured residuals after accounting for known contributions under a chosen baseline contract. LTT provides a disciplined method to quantify and structure such residuals.

Hypothesis 11.2 (Void-ledger interpretation (optional, separable))**.** If residual-ledger accumulation exhibits persistent, cross-domain structure (laboratory closure failures, astrophysical missing-component maps, and detector residual coherence) that cannot be removed by improving the baseline contract, then the Void ledger may correspond to a real physical degree of freedom. This statement is falsifiable only via repeated closure/intervention audits across domains.

Remark 11.3 (Separation principle)*.* The LTT is valid as an operational residual tool even if the Void-ledger interpretation is false. This separation is intentional.

A Symbol Summary (Supplement)

Symbol	Meaning
$\mathcal{X}$	Space substrate (domain of matter positions)
Σ_X, Σ_T	Matter-side and energy/field-side reference systems
$\mathcal{O}$	Common observable space $\mathcal{O} \subseteq \mathbb{R}^n$ ($\Sigma_{\text{common}} \equiv \mathcal{O}$)
Π_X, Π_T	Projections into $\mathcal{O}$ from Σ_X, Σ_T
r	Residual vector $r = \Pi_X(A) - \Pi_T(B)$
$\mathbf{\Sigma}$	Covariance of measurement + model uncertainty in $\mathcal{O}$
$\mathbf{W}$	Weight matrix (often $\mathbf{\Sigma}^{-1}$)
δ	Calibrated mismatch $\delta = \sqrt{r^\top \mathbf{W} r}$ (or likelihood distance)
$\mathbf{L}$	Loss-channel vector $\mathbf{L} = \mathcal{C}(r)$
$\mathcal{C}$	Channelization map (residual $\to$ channels)
$\mathcal{V}_E, \mathcal{V}_I$	Energetic and informational Void ledgers
V_E, V_I	Accumulated energetic / informational ledgers over time